Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/17/2004
PI Name:	Walsworth, Ronald L. Ph.D.		
Project Title:	Advanced Technology Development of Laser-Polarized Noble Gas Magnetic Resonance		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	08/01/2003	**End Date:**	09/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1489		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are continuing development of laser-polarized noble gas nuclear magnetic resonance (NMR) as a novel biomedical imaging tool for ground-based and possibly space-based application. This emerging multidisciplinary technology enables high-resolution gas-space magnetic resonance imaging (MRI), e.g., of the lung, as well as studies of tissue perfusion. In addition, laser-polarized noble gases (3He and 129Xe) do not require a large magnetic field for sensitive detection, which opens the door to practical MRI at very low magnetic fields with an open-geometry device. We are pursuing two specific aims in the current project: (i) development of an optimized, open-geometry, low-field MRI system for human lung imaging; and (ii) development of minimally-invasive NMR technologies to probe acinar structure, gas exchange, alveolar oxygen concentration, and regional blood volume in human and animal lungs. During the reporting year, the task objectives were: (i) design and begin construction of an optimized, second-generation system for open-geometry low-field human MRI; (ii) develop the capability for high-field MRI studies of human lungs,		

	to enable studies of acinar structure, gas exchange, alveolar oxygen concentration, and regional blood volume; and (iii) continue related technique-demonstration studies in model systems. All of the tasks being pursued during the reporting year have an interrelated significance: to realize a powerful, minimally-invasive technology for imaging pulmonary function, with a wide range of bioastronautics and Earth-based applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The techniques under development (including low-field MRI and gas-diffusion NMR of inhaled, laser-polarized noble gas) will provide novel tools for furthering understanding of cardiopulmonary physiology, and will serve as non-invasive diagnostics of, and treatment monitors for, lung diseases such as asthma and emphysema. Also, the open-geometry low-field MRI system will be applicable to patient groups for which traditional high-field MRI is prohibited or difficult: e.g., patients with implants or claustrophobia, or with size or physical constraints that prevent being placed in the tight tube of a traditional MRI instrument.
Task Progress:	* Completed design for optimized, second-generation low-field human MRI system. * Began construction of second-generation low-field human MRI system, located at the Harvard-Smithsonian Center for Astrophysics: main field coils, gradient coils, mounting frame, RF-shielded room, and control console were all constructed and/or installed. * Developed hardware and software for high-field human lung MRI, to be performed with polarized 3He gas at the Brigham and Women's Hospital. * Demonstrated simultaneous NMR measurements of coherent and diffusive gas flow in model porous systems (a technique with applications to lung physiology).
Bibliography Type:	Description: (Last Updated: 07/25/2019)
Abstracts for Journals and Proceedings	Wang, R.; Rosen, M.S.; Candela, D.; Mair, R. W.; Walsworth, R. L. "Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe" Seventh International Meeting on Recent Advances in MR Applications to Porous Media, Paris, France, July 2004. Conference Proceedings, pg. 36. , Jul-2004
Abstracts for Journals and Proceedings	Rosen, M.S.; Tsai, L. L.; Mair, R. W.; Walsworth, R. L. "Design of an optimized open-access human-scale MRI magnet for orientational lung study." Twelfth Scientific Meeting of the International Society for Magnetic Resonance in Medicine, Kyoto, Japan, May, 2004. Conference Proceedings, pg. 749. , May-2004
Abstracts for Journals and Proceedings	Mair, R.W.; Hrovat, M. I.; Patz, S.; Rosen, M.S.; Ruset, I.; Topulos, G. P.; Tsai, L. L.; Hersman, F. W.; Walsworth, R. L. "Initial Orientational Lung Imaging in an Open-Access Human-Scale Low-Field MRI System" Twelfth Scientific Meeting of the International Society for Magnetic Resonance in Medicine, Kyoto, Japan, May, 2004. Conference Proceedings, pg. 859. , May-2004
Abstracts for Journals and Proceedings	Pavlin, T.; Topulos, G. P.; Butler, J.P.; Mair, R. W.; Walsworth, R. L.; Patz, S. "Importance of Susceptibility Effects in the Determination of Fractional Regional Blood Volume (fRBV)." Twelfth Scientific Meeting of the International Society for Magnetic Resonance in Medicine, Kyoto, Japan, May, 2004. Conference Proceedings, pg. 2183. , May-2004
Articles in Peer-reviewed Journals	Mair, R.W.; Hrovat, M.I.; Patz, S.; Rosen, M.S.; Ruset, I.C.; Topulos, G.P.; Tsai, L.L.; Butler, J.P.; Hersman, F.W.; Walsworth, R.L. "3He Lung Imaging in an Open Access, Very-Low-Field Human MRI System" Accepted for publication in Magnetic Resonance in Medicine , Dec-2004
Articles in Peer-reviewed Journals	Wang, R.; Rosen, M.S.; Candela, D.; Mair, R.W.; Walsworth, R.L. "Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe Study of Gas-Fluidization Dynamics with Laser-Polarized 129Xe" Accepted for publication in Magnetic Resonance Imaging. , Sep-2004